Analog Buck-Boost (LTC3440, 3.9 V): Working Document

Digital Audio Player • Rev 10

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Contents

1. Summary 1

2. Design Requirements 1

3. Pin Configuration 2

4. External Components 2

5. Equations and Results 2

6. Important Values 3

7. Open Problems 3

8. Revision Log 3

9. Sources 4

1. Summary

Field	Value
Manufacturer / MPN	Analog Devices, LTC3440EMS#TRPBF (#1-050), U200
Package	10-lead MSOP (MS)
Function	Synchronous buck-boost, SYS → 3.9 V intermediate analog rail

2. Design Requirements

Requirement	Value
Input	SYS, 3.0 to 4.2 V
Output	3.9 V
Load	Analog chain only: two LT3042 (+3V3_ANALOG, +3V3_DAC) and LT3462A → LT3093
Output current	≈ 263 mA peak: U204 104 + DAC LDO 10 (8 load + ≈ 2 quiescent) + U204 quiescent ≈ 2 + LT3462A input 147; 250 mA design current exceeded by ≈ 5 %
Fault ceiling	≈ 0.31 A (U204 at 125 mA + DAC LDO + LT3462A at its switch limit)
Switching frequency	1.2 MHz, fixed (Burst Mode off)
Output ripple target	0.5 % (19.5 mV pp)
Soft-start	Output up in ≈ 6 ms, MCU-enabled

Requirement	Value
Constraints	Low noise · Small footprint

Source: [3]

3. Pin Configuration

Pin	Name	Connection
1	RT	R_T to GND
2	MODE/SYNC	GND (fixed frequency)
3, 4	SW1, SW2	L
5	GND	GND
6	VOUT	+3V9, C_OUT, R1
7	VIN	SYS, C_IN
8	SHDN/SS	R_SS from GPIO35 (1 MΩ pull-down on the GPIO side), C_SS to GND
9	FB	R1 / R2 midpoint
10	VC	C_P1 and R_Z to FB

4. External Components

Ref	Value / Rating	MPN	AL-PN
L (L200)	6.8 µH ±20 %, I_sat 3.5 A, 74.1 mΩ	Bourns SRP4030FA-6R8M	#1-076
C_IN (C203)	10 µF 10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
C_OUT (C204)	22 µF 25 V X7S 1210	Murata GCM32EC71E226KE36L	#1-073
R1 (R200)	220 kΩ 0.1 % 0402	YAGEO RC0402BR-07220KL	#1-074
R2 (R212)	100 kΩ 0.1 % 0402	YAGEO RT0402BRD07100KL	#1-131
R_T (R207)	49.9 kΩ 0.1 % 0402	YAGEO RT0402BRD0749K9L	#1-138
R_Z (R213)	15 kΩ 0.1 % 0402	YAGEO RT0402BRD0715KL	#1-078
C_P1 (C212)	1.5 nF 2 % C0G 0402	Murata GRM1555C1H152GA01D	#1-077
R_SS (R203)	54.9 kΩ 0.1 % 0402	YAGEO RT0402BRD0754K9L	#1-079
C_SS (C213)	0.1 µF 50 V X7R 0402	Murata GCM155R71H104KE02D	#1-080
R_PD (R308, MCU sheet)	1 MΩ 0.1 % 0402	YAGEO RE0402BRE071ML	#1-123

Source: [2]

5. Equations and Results

$$R_T = 6 \times 10^{10} / 1.2\,MHz = 50\,k\Omega; \quad V_{OUT} = 1.22\,(1 + 220/100) = 3.904\,V \tag{1}$$

$$Ripple_{boost} = \frac{3.0 \times 0.9}{1.2 \times 0.25 \times 6.8 \times 3.9} = 0.34\ (85\,mA_{pp}); \quad Ripple_{buck} = 0.14 \tag{2}$$

$$\frac{\Delta V_{OUT}}{V_{OUT}} = \frac{0.263 \times 0.9}{22 \times 3.9^2 \times 1.2} = 0.059\,\%\ (2.3\,mV_{pp}); \quad buck\ \ 0.004\,\% \tag{3}$$

$$I_{L,\,pk} = \frac{0.263 \times 3.9}{3.0 \times 0.85} + \frac{0.085}{2} = 0.44\,A \tag{4}$$

$$\tau = R_{SS}C_{SS} = 5.49\,ms:\ \ enable\ (1.0\,V)\ 2.0\,ms,\ \ ramp\ complete\ (2.2\,V)\ 6.0\,ms \tag{5}$$

$$V_{IN}: V_{OUT}(1 - 150\,ns \cdot f)\ to\ V_{OUT}/(1 - 150\,ns \cdot f) = 3.20\ to\ 4.76\,V \tag{6}$$

Source: [1]

Case	U204	DAC LDO	LT3462A input	LTC3440 load
Quiescent	≈ 34 mA	≈ 10 mA	≈ 38 mA	≈ 82 mA
Full scale, 23.8 Ω	≈ 106 mA	≈ 10 mA	≈ 147 mA	≈ 263 mA
Fault	125 mA	≈ 10 mA	≈ 0.18 A	≈ 0.31 A

U204 figures include its ≈ 2 mA ground current. Previous revisions omitted the DAC oscillators (≤ 5.5 mA) from the +3V3_ANALOG load.

6. Important Values

Item	Value
Switch current limit	2.7 A typ (2.5 to 2.9 A)
Calculated output ripple	2.3 mV pp boost (3.0 V in), 0.16 mV pp buck (4.2 V in); nominal C_OUT
Simulated output ripple	7.7 mV pp (target 19.5 mV)
Load step 150 ↔ 250 mA	±90 mV, no ringing, 0.3 ms recovery
Minimum rail during step	3.74 V
Fault ceiling	≈ 0.31 A: inductor peak ≈ 0.5 A vs 3.5 A I_sat

7. Open Problems

None.

8. Revision Log

Rev	Date	Change
0 to 4	2026-09-21 to 23	Initial design, simulation, load updates, inverter branch.
5 to 7	2026-09-25 to 27	Reformatted; LT3462A efficiency; load from Power Budget Rev 6 (253 mA).
8	2026-10-03	Second LT3042 (DAC rail) and oscillator load added: ≈ 263 mA peak. C_IN 10 µF (#1-081); R2 0402 (#1-131). Designators added. Added: load margin item.
9	2026-10-03	DAC rail named +3V3_DAC; load unchanged.
10	2026-10-16	Added calculated output voltage ripple, equation (3): 2.3 mV pp boost, 0.16 mV pp buck at nominal C_OUT. Later equations renumbered. Datasheet link added to [1]. Closed: load margin item (author). R_T 49.9 kΩ 0402 (#1-138), per schematic.

9. Sources

1. LTC3440 Product Page and Datasheet (Analog Devices)
2. DAP schematic sheet DAP-Power-2, 3.9 V buck-boost (author, 2026-10-16)
3. Power Budget, Rev 7